LAB EXPERIMENTS

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Experiment – 1 :

```
# Create DataFrame
data = {
  "DEPARTMENT_ID": [10, 20, 30, 40, 50, 60, 70, 80, 90, 100,
                    110, 120, 130, 140, 150, 160, 170, 180,
                    190, 200, 210, 220, 230, 240, 250, 260, 270],
  "DEPARTMENT_NAME": [
    "Administration", "Marketing", "Purchasing", "Human Resources",
    "Shipping", "IT", "Public Relations", "Sales", "Executive",
    "Finance", "Accounting", "Treasury", "Corporate Tax",
    "Control And Credit", "Shareholder Services", "Benefits",
    "Manufacturing", "Construction", "Contracting", "Operations",
    "IT Support", "NOC", "IT Helpdesk", "Government Sales",
    "Retail Sales", "Recruiting", "Payroll"
  ],
  "MANAGER_ID": [200, 201, 114, 203, 121, 103, 204, 145, 100,
                 108, 205, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
  "LOCATION_ID": [1700, 1800, 1700, 2400, 1500, 1400, 2700, 2500, 1700,
                  1700, 1700, 1700, 1700, 1700, 1700, 1700, 1700, 1700,
                  1700, 1700, 1700, 1700, 1700, 1700, 1700, 1700, 1700]
}
```

Create DataFrame

df = pd.DataFrame(data)

Display DataFrame

print("Employees Data:")

print(df)

Select distinct Department IDs

print("\nDistinct Department IDs:")

print(df["DEPARTMENT_ID"].drop_duplicates())

```
Distinct Department IDs:
0      10
1      20
2      30
3      40
4      50
5      60
6      70
7      80
8      90
9     100
10    110
11    120
12    130
13    140
14    150
15    160
16    170
17    180
18    190
19    200
20    210
21    220
22    230
23    240
24    250
25    260
26    270
Name: DEPARTMENT_ID, dtype: int64
```

Experiment – 2 :

```python
import pandas as pd

# Create DataFrame
data = {
    "EMPLOYEE_ID": [102, 101, 101, 201, 114, 122, 200, 176, 176, 200],
    "START_DATE": ["2001-01-13", "1997-09-21", "2001-10-28", "2004-02-17",
                   "2006-03-24", "2007-01-01", "1995-09-17", "2006-03-24",
                   "2007-01-01", "2002-07-01"],
    "END_DATE": ["2006-07-24", "2001-10-27", "2005-03-15", "2007-12-19",
                 "2007-12-31", "2007-12-31", "2001-06-17", "2006-12-31",
                 "2007-12-31", "2006-12-31"],
    "JOB_ID": ["IT_PROG", "AC_ACCOUNT", "AC_MGR", "MK_REP",
               "ST_CLERK", "ST_CLERK", "AD_ASST", "SA_REP",
               "SA_MAN", "AC_ACCOUNT"],
    "DEPARTMENT_ID": [60, 110, 110, 20, 50, 50, 90, 80, 80, 90]
}

df = pd.DataFrame(data)

# Employees with two or more jobs
result = df.groupby("EMPLOYEE_ID").filter(lambda x: len(x) >= 2)

print("Employees who have done two or more jobs:")
print(result["EMPLOYEE_ID"].drop_duplicates())
```

```python
# Create DataFrame
data = {
    "EMPLOYEE_ID": [102, 101, 101, 201, 114, 122, 200, 176, 176, 200],
    "START_DATE": ["2001-01-13", "1997-09-21", "2001-10-28", "2004-02-17",
                   "2006-03-24", "2007-01-01", "1995-09-17", "2006-03-24",
                   "2007-01-01", "2002-07-01"],
    "END_DATE": ["2006-07-24", "2001-10-27", "2005-03-15", "2007-12-19",
                 "2007-12-31", "2007-12-31", "2001-06-17", "2006-12-31",
                 "2007-12-31", "2006-12-31"],
    "JOB_ID": ["IT_PROG", "AC_ACCOUNT", "AC_MGR", "MK_REP",
               "ST_CLERK", "ST_CLERK", "AD_ASST", "SA_REP",
               "SA_MAN", "AC_ACCOUNT"],
    "DEPARTMENT_ID": [60, 110, 110, 20, 50, 50, 90, 80, 80, 90]
}

df = pd.DataFrame(data)

# Employees with two or more jobs
result = df.groupby("EMPLOYEE_ID").filter(lambda x: len(x) >= 2)

print("Employees who have done two or more jobs:")
print(result["EMPLOYEE_ID"].drop_duplicates())
```

```
Employees who have done two or more jobs:
1    101
6    200
7    176
Name: EMPLOYEE_ID, dtype: int64
```

Experiment – 3 :

```
import pandas as pd

# Create DataFrame
data = {
   "JOB_ID": ["AD_PRES","AD_VP","AD_ASST","FI_MGR","FI_ACCOUNT","AC_MGR",
           "AC_ACCOUNT","SA_MAN","SA_REP","PU_MAN","PU_CLERK",
           "ST_MAN","ST_CLERK","SH_CLERK","IT_PROG","MK_MAN",
           "MK_REP","HR_REP","PR_REP"],

   "JOB_TITLE": ["President","Administration Vice President",
             "Administration Assistant","Finance Manager",
             "Accountant","Accounting Manager","Public Accountant",
             "Sales Manager","Sales Representative",
             "Purchasing Manager","Purchasing Clerk",
             "Stock Manager","Stock Clerk","Shipping Clerk",
             "Programmer","Marketing Manager",
             "Marketing Representative",
             "Human Resources Representative",
             "Public Relations Representative"],

   "MIN_SALARY": [20080,15000,3000,8200,4200,8200,4200,10000,6000,
             8000,2500,5500,2008,2500,4000,9000,4000,4000,4500],

   "MAX_SALARY": [40000,30000,6000,16000,9000,16000,9000,20080,12008,
             15000,5500,8500,5000,5500,10000,15000,9000,9000,10500]
}

df = pd.DataFrame(data)
```

Sort by JOB_TITLE in descending order

result = df.sort_values(by="JOB_TITLE", ascending=False)

print(result)

```python
df = pd.DataFrame(data)

# Sort by JOB_TITLE in descending order
result = df.sort_values(by="JOB_TITLE", ascending=False)

print(result)
```

```
        JOB_ID                         JOB_TITLE  MIN_SALARY  MAX_SALARY
11      ST_MAN                     Stock Manager        5500        8500
12    ST_CLERK                       Stock Clerk        2008        5000
13    SH_CLERK                    Shipping Clerk        2500        5500
8       SA_REP              Sales Representative        6000       12008
7       SA_MAN                     Sales Manager       10000       20080
9       PU_MAN                Purchasing Manager        8000       15000
10    PU_CLERK                  Purchasing Clerk        2500        5500
18      PR_REP   Public Relations Representative        4500       10500
6   AC_ACCOUNT                 Public Accountant        4200        9000
14     IT_PROG                        Programmer        4000       10000
0      AD_PRES                         President       20080       40000
16      MK_REP          Marketing Representative        4000        9000
15      MK_MAN                 Marketing Manager        9000       15000
17      HR_REP   Human Resources Representative        4000        9000
3       FI_MGR                   Finance Manager        8200       16000
1        AD_VP     Administration Vice President       15000       30000
2      AD_ASST          Administration Assistant        3000        6000
5       AC_MGR                Accounting Manager        8200       16000
4   FI_ACCOUNT                        Accountant        4200        9000
```